

CHAPTER 9

General Equilibrium

In Chapters 6-8 we have studied the decisions of households and firms in isolation. In this chapter we look at how they all fit together.

In microeconomics we say that there's an equilibrium in a competitive market for some good if supply equals demand: everyone buys or sells as much as they want and the outcome is that sales equal purchases. General equilibrium is the same idea but applied to many goods at once.

The model economy we'll be looking at will have:

- a representative household making consumption-savings-leisure decisions
- a representative firm choosing to hire labor and rent capital
- a representative investment firm carrying out investment

We'll first look at this in a simplified two-period model and then we'll extend the analysis to an infinite-period case.

9.1 General Equilibrium in a Two-Period Economy

We are going to assume that the economy lasts only for two periods. In the first period, there is already some level of capital K_1 in place, and it's owned by the representative household. Each period, the production function is $F(K, L)$ and there are competitive markets for labor and renting capital. The initial capital stock depreciates at rate δ between period 1 and period 2, and it's possible to invest in order to build more capital. After period 2 it's the end of the world, so capital depreciates completely ($\delta_2 = 1$) and there is no more investment.

Household, Firm and Investment Firm Problems

The representative household must choose how much to consume and also how much of its time to dedicate to leisure and consumption. It solves a problem like the one we looked at in Section 7.4:

$$\begin{aligned} \max_{c_1, l_1, c_2, l_2} \quad & u(c_1) + v(l_1) + \beta [u(c_2) + v(l_2)] \\ \text{s.t.} \quad & \end{aligned} \tag{9.1.1}$$

$$c_1 + \frac{1}{1+r}c_2 \leq w_1(1-l_1) + \frac{1}{1+r}w_2(1-l_2) + K_1(1+r_1^K - \delta) + \Pi$$

The only thing that is different between this problem and the one we know from Section 7.4 is that the household starts with some initial wealth: it owns the initial capital stock and earns a rental (net of depreciation) for it in the first period. Also, the household owns the firms, so if they were to make profits, the present value of those profits, which we denote by Π , would be part of the household's budget. Π is given by $\Pi = \Pi_1^F + \frac{\Pi_2^F}{1+r} + \Pi^I$, where Π_t^F are the profits of productive firms in period t and Π^I are the profits of investment firms. In equilibrium, it will be the case that $\Pi = 0$, so we can just ignore this part.

There is a representative firm. In each period, the firm solves the profit maximization problem we saw in Chapter 4:

$$\Pi_t^F = \max_{K, L} F(K, L) - w_t L - r_t^K K$$

Investment consists of building period-2 capital. We are going to imagine that there is a representative investment firm that carries out all the investment. This firm buys the $(1-\delta)K_1$ units of used period-1 capital, adds I units of investment and obtains $K_2 = (1-\delta)K_1 + I$ units of period-2 capital (here we are assuming there are no adjustment costs like the ones we had in Section 8.1). The investment firms then rents out the K_2 units of capital at the rental rate r_2^K (given that the second period is the end of the world, they *fully* depreciate in period 2). Converted back to present value, this means the investment firm earns $\frac{r_2^K}{1+r}K_2$ from renting out the capital. The problem of the investment firm is:

$$\Pi^I = \max_I \underbrace{\frac{r_2^K}{1+r} [(1-\delta)K_1 + I]}_{\text{Present Value of } K_2 \text{ rental}} - \underbrace{[(1-\delta)K_1 + I]}_{\text{Cost of } K_2} \tag{9.1.2}$$

Equilibrium Definition

Definition 9.1.

Given an initial K_1 , a competitive equilibrium consists of:

1. An allocation $\{c_1, l_1, c_2, l_2, I, K_2, L_1, L_2\}$.
2. Prices $\{w_1, w_2, r_1^K, r_2^K, r\}$.

such that:

1. $\{c_1, l_1, c_2, l_2\}$ solves the household's problem, taking prices as given.
2. L_t, K_t solve the firm's problem for $t = 1, 2$, taking prices as given.

3. I solves the investment firm's problem, taking prices as given.

4. Markets for goods and labor in each period clear:

(a) Goods:

$$\underbrace{F(K_1, L_1)}_{\text{GDP}} = \underbrace{c_1}_{\text{Consumption}} + \underbrace{I}_{\text{Investment}} \quad (9.1.3)$$

$$\underbrace{F(K_2, L_2)}_{\text{GDP}} = \underbrace{c_2}_{\text{Consumption}} \quad (9.1.4)$$

(b) Capital:

$$K_2 = K_1(1 - \delta) + I \quad (9.1.5)$$

(c) Labor:

$$L_t + l_t = 1 \quad (9.1.6)$$

A competitive equilibrium satisfies two basic properties:

- Everyone is, individually, making the best choices they can. This is represented by conditions 1-3, which say that each individual household and each individual firm makes its choices in their own best interest.
- Things add up, i.e. everyone's choices are consistent with everyone else's choices. This is represented in condition 4. Condition 4(a) says that all the output in the economy is used for either consumption or investment in the first period and, since it's the end of the world, only for consumption in the second period. (Recall that this is a closed economy with no government, so there is no other use for output). Condition 4(b) says that the capital that firms want to use in period 2 is equal to the amount of period-1 capital that remains plus the amount that investment firms chose to build. Condition 4(c) says that the labor that firms choose to hire (L_t) plus the amount of time the households choose to dedicate to leisure (l_t) add up to the entire amount of time available, which is normalized to 1.

Describing an Equilibrium

In this section we'll find a system of equations whose solution represents the economy's competitive equilibrium. For now we are going to leave it as a mathematical expression and the economics it contains might be a little hard to discern. We'll use these equations to think more about economics in later chapters.

We know from Section 7.4 that the first order conditions are (7.4.2) and (7.4.3), which we just restate here:

$$\frac{v'(l_t)}{u'(c_t)} = w_t \quad (9.1.7)$$

$$u'(c_1) = \beta(1 + r)u'(c_2) \quad (9.1.8)$$

Equation (9.1.7) is the same as (7.4.2) and it describes how the household trades off leisure and consumption. Equation (9.1.8) is the same as (6.3.9) and (7.4.3) and it describes how the household trades off present and future consumption.

As we saw in Chapter 4, the solution to the firm's problem can be summarized by the first-order conditions:

$$F_K(K_t, L_t) - r_t^K = 0 \quad (9.1.9)$$

$$F_L(K_t, L_t) - w_t = 0 \quad (9.1.10)$$

(which are the same as (4.4.1) and (4.4.2)).

The investment firm's problem (9.1.2) is linear in I . This means that unless $1 + r = r_2^K$, investment firms would be able to make infinite profits by choosing either $I = \infty$ or $I = -\infty$ depending on which way the inequality goes. This would be inconsistent with capital-market clearing. Therefore it must be that in equilibrium:

$$1 + r = r_2^K \quad (9.1.11)$$

which is just equation (4.4.12) when depreciation is set to 1. Recalling the definition of net present value from Chapter 8, this says that in equilibrium the NPV of investment must be zero.

Replacing (9.1.10) into (9.1.7) we obtain:

$$\underbrace{\frac{v'(l_t)}{u'(c_t)}}_{\text{Marginal Rate of Substitution}} = \underbrace{F_L(K_t, L_t)}_{\text{Marginal Rate of Transformation}} \quad (9.1.12)$$

Equation (9.1.12) summarizes how this economy will allocate the use of time. On the left hand side, the expression $\frac{v'(l_t)}{u'(c_t)}$ describes how the representative household is willing to trade off leisure against consumption. On the right hand side, $F_L(K_t, L_t)$ describes how the available technology is able (at the margin) to convert time into output.

Replacing (9.1.11) and (9.1.9) into (9.1.8) and we obtain:

$$\underbrace{\frac{u'(c_1)}{\beta u'(c_2)}}_{\text{Marginal Rate of Substitution}} = \underbrace{F_K(K_2, L_2)}_{\text{Marginal Rate of Transformation}} \quad (9.1.13)$$

Equation (9.1.13) summarizes how this economy will allocate output between the present and the future. On the left hand side, the expression $\frac{u'(c_1)}{\beta u'(c_2)}$ describes how the representative household is willing to trade off present consumption against future consumption. On the right hand side, $F_K(K_2, L_2)$ describes how the available technology is able, by building a marginal unit of capital, to convert current output into future output.

Equations (9.1.12) and (9.1.13) summarize what the general equilibrium will look like. We'll come back to them many times.

9.2 The First Welfare Theorem

A competitive economy will allocate resources in one specific way. What if we could choose a different allocation? Would that be a good idea?

In order to answer this question it is convenient to invoke the metaphor of a “social planner”. The idea is to imagine that, instead of making their decisions individually, everyone delegates decisions to a benevolent social planner. Is it the case that the social planner would want to change the allocation of resources to something other than the competitive equilibrium? We’ll show that, under the assumptions we’ve made so far, the answer is “no”. Even a social planner that was perfectly benevolent and had no practical, political or cognitive difficulties in choosing among all the possible allocations of resources would be satisfied with the outcome of competitive markets.

The fictitious social planner solves the following problem:

$$\begin{aligned}
 & \max_{c_1, l_1, c_2, l_2, L_1, L_2, K_2} u(c_1) + v(l_1) + \beta [u(c_2) + v(l_2)] \\
 & \quad s.t. \\
 & K_2 \leq (1 - \delta) K_1 + F(K_1, L_1) - c_1 \\
 & c_2 \leq F(K_2, L_2) \\
 & L_t \leq 1 - l_t \quad \text{for } t = 1, 2 \\
 & K_1 \text{ given}
 \end{aligned} \tag{9.2.1}$$

What does this optimization problem represent, in economic terms? First, the planner indeed is benevolent. Its objective function is the same as the objective of the representative household: the planner wants to make the representative household happy. Second, the planner is quite powerful: it can tell everyone exactly how much to work, consume and invest without worrying about whether they will obey the instructions. In that regard, our fictitious social planner is much more powerful than a government could possibly be. Lastly, the planner is not all-powerful: it is constrained by the technological possibilities of the economy. The constraints, which are identical to conditions (9.1.3)-(9.1.6), say that in order to accumulate capital it is necessary to give up consumption and in order to produce output it is necessary to use labor, which requires giving up leisure. These constraints are not exactly budget constraints since the planner is not buying or selling from anyone; rather, they are technological constraints.

Let’s replace the constraints in the objective function and then solve the planner’s problem. Replace $c_1 = (1 - \delta) K_1 + F(K_1, L_1) - K_2$, $c_2 = F(K_2, L_2)$ and $l_t = 1 - L_t$ into the objective to get:¹

$$\max_{L_1, L_2, K_2} u((1 - \delta) K_1 + F(K_1, L_1) - K_2) + v(1 - L_1) + \beta [u(F(K_2, L_2)) + v(1 - L_2)]$$

The first order conditions with respect to L_1, L_2 and K_2 respectively are:

$$\begin{aligned}
 u'((1 - \delta) K_1 + F(K_1, L_1) - K_2) F_L(K_1, L_1) - v'(1 - L_1) &= 0 \\
 \beta [u'(F(K_2, L_2)) F_L(K_2, L_2) - v'(1 - L_2)] &= 0 \\
 -u'((1 - \delta) K_1 + F(K_1, L_1) - K_2) + \beta u'(F(K_2, L_2)) F_K(K_2, L_2) &= 0
 \end{aligned}$$

which simplify back to (9.1.12) and (9.1.13), which are the equations that describe the competitive equilibrium.

¹Alternatively, we can set up a Lagrangian with all the constraints.

Therefore the equations that define the solution to the social planner's problem are the same as those which define the competitive equilibrium! This is not a coincidence. The following result proves that the allocations that result from a competitive equilibrium must achieve the optimum in the planner's problem.

Proposition 9.1 (First Welfare Theorem). If an allocation $\{c_1, l_1, c_2, l_2, I, K_2, L_1, L_2\}$ is part of a competitive equilibrium then it solves the social planner's problem.

Proof. Proceed by contradiction. Suppose that an allocation $\{c_1, l_1, c_2, l_2, I, K_2, L_1, L_2\}$ with prices $\{w_1, w_2, r_1^K, r_2^K, r\}$ is a competitive equilibrium but there is another allocation $\{\hat{c}_1, \hat{l}_1, \hat{c}_2, \hat{l}_2, \hat{I}, \hat{K}_2, \hat{L}_1, \hat{L}_2\}$ that satisfies the constraints on the social planner's problem but achieves strictly higher utility for the household. This implies that:

$$\hat{c}_1 + \frac{1}{1+r}\hat{c}_2 > w_1(1 - \hat{l}_1) + \frac{1}{1+r}w_2(1 - \hat{l}_2) + K_1(1 + r_1^K - \delta) + \Pi \quad (9.2.2)$$

Equation (9.2.2) says, if prices are the equilibrium prices, then the household cannot afford the consumption-leisure combination $\{\hat{c}_1, \hat{l}_1, \hat{c}_2, \hat{l}_2\}$. Otherwise, since it's presumed to be strictly better, the household would have chosen it. Furthermore, the fact that in equilibrium firms and investment firms are maximizing profits implies that by choosing the equilibrium quantities they make at least as much profit as they would by choosing the alternative quantities:

$$\begin{aligned} \Pi_1^F &= F(K_1, L_1) - w_1 L_1 - r_1^K K_1 \geq F(K_1, \hat{L}_1) - w_1 \hat{L}_1 - r_1^K K_1 \\ \Pi_2^F &= F(K_2, L_2) - w_2 L_2 - r_2^K K_2 \geq F(\hat{K}_2, \hat{L}_1) - w_2 \hat{L}_1 - r_2^K \hat{K}_2 \\ \Pi^I &= \left(\frac{r_2^K}{1+r} - 1\right) [(1-\delta)K_1 + \hat{I}] \geq \left(\frac{r_2^K}{1+r} - 1\right) [(1-\delta)K_1 + \hat{I}] \end{aligned}$$

Therefore, using the definition:

$$\Pi \equiv \Pi_1^F + \frac{\Pi_2^F}{1+r} + \Pi^I$$

to add the profits of production firms and investment firms, we have that:

$$\Pi \geq F(K_1, \hat{L}_1) - w_1 \hat{L}_1 - r_1^K K_1 + \frac{F(\hat{K}_2, \hat{L}_1) - w_2 \hat{L}_1 - r_2^K \hat{K}_2}{1+r} + \left(\frac{r_2^K}{1+r} - 1\right) [(1-\delta)K_1 + \hat{I}] \quad (9.2.3)$$

Replacing (9.2.3), $\hat{L}_t \leq 1 - \hat{l}_t$ and $\hat{I} = K_2 - (1-\delta)K_1$ into (9.2.2) and simplifying, we obtain:

$$\hat{c}_1 + K_2 + \frac{1}{1+r}\hat{c}_2 > F(K_1, \hat{L}_1) + (1-\delta)K_1 \frac{F(\hat{K}_2, \hat{L}_1)}{1+r}$$

This implies that one of the following statements must be true. Either:

$$\hat{c}_1 + K_2 > F(K_1, \hat{L}_1) + (1-\delta)K_1$$

or

$$\hat{c}_2 > F(\hat{K}_2, \hat{L}_1)$$

either of which contradicts the assumption that $\{\hat{c}_1, \hat{l}_1, \hat{c}_2, \hat{l}_2, \hat{I}, \hat{K}_2, \hat{L}_1, \hat{L}_2\}$ satisfies the constraints on the planner's problem. $\square$

What is the economic logic behind the First Welfare Theorem? Why are the social planner's choices the same as those the market produces? The social planner chooses an allocation to maximize utility subject to technological possibilities. Conversely, in a competitive economy, each household maximizes utility subject to prices (i.e. wages, the rental rate of capital and the interest rate), as equations (9.1.7) and (9.1.8) indicate; however, those prices in turn reflect technological possibilities, as equations (9.1.9) and (9.1.10) indicate. Therefore the household is also, indirectly, maximizing utility subject to technological possibilities.

What we have shown is really just a special case of the First Welfare Theorem. The result is much more general. In particular, it is still true if:

- There are many different goods at each date (e.g. apples, oranges, etc.) instead of just a general consumption good.
- There is (exogenous) technological progress.
- There is uncertainty, as long as there are “complete markets”, i.e. markets to trade insurance against every possible state of the world, at competitive prices.
- There are many different households with different preferences, different abilities and different wealth instead of a representative household. However, for this case the theorem needs to be stated slightly differently. If an economy has different households, there is no unique way to define the social planner's problem because the planner would have to decide how much the utility of each different household matters. Hence, the more general version of the FWT says that if an allocation is part of a competitive equilibrium then it is Pareto optimal: there is no technologically feasible way to make someone better off without making someone else worse off.

The FWT is an extremely useful guide for thinking about public policy. In an economy where the conditions for the theorem hold, then *any* policy that makes a household better off, relative to the competitive equilibrium, necessarily makes someone else worse off. Does this mean that no policy is ever justified? Some people interpret the theorem to imply just that, but this conclusion requires an extra bit of political philosophy. Depending on one's views on the nature of justice, it is quite possible to advocate for a policy that benefits some groups of people at the expense of others (for instance, policies that benefit the poor at the expense of the rich or the old at the expense of the young). What the FWT does clarify is that, if the economy is competitive, the *only* possible economic justification for a policy is that one views the resulting redistribution as desirable.

The conditions for the FWT are quite strict and no one believes that they hold exactly in practice. Two of the main things that would make the FWT *not* hold are:²

²Other features that would make the FWT fail include asymmetric information, incomplete markets and borrowing constraints.

- Monopoly power. Monopolists reduce quantity, relative to a competitive producer, in order to raise prices. Therefore it is not true that they equate the value of the marginal product to the factor prices, which is the key step that leads to equations (9.1.12) and (9.1.13). Hence, the FWT does not hold. In Chapter 14 we study a model that features monopoly power.
- Externalities. In an economy with externalities, the private value and the social value of a good do not coincide. If I decide to hire a (good) orchestra to play in my garden and all my neighbors enjoy it, the social value this produces exceeds the private value that I obtain. When I decide whether or not to hire the orchestra, I will ignore the benefit to my neighbors while a benevolent social planner would take it into account, so I will hire an orchestra less often than what the social planner would want. Similarly, I will over-pollute, under-invest in preventing communicable diseases, etc.

Again, the FWT helps to organize the analysis of public policy. If a policy is not about redistribution, then it can only be justified economically on the basis of which failure of the FWT it's designed to address.³ For instance, zoning regulations can (perhaps) be justified as a way to deal with externalities: if I could build a tall building next to your house, I would make your garden less sunny. Antitrust policies can (perhaps) be justified as a way to deal with monopoly power. It is often useful to begin thinking about a policy problem by asking what is the failure of the FWT that the policy is designed to address.

9.3 General Equilibrium in an Infinite-Period Economy

Let's now look at what happens in an economy that lasts infinite periods. We'll proceed in the same way as before: describe the household's problem, the firm's problem, the investment decision and finally define and describe an equilibrium.

The representative household solves the following problem:

$$\begin{aligned} \max_{c_t, l_t, a_{t+1}} \sum_{t=0}^{\infty} \beta^t [u(c_t) + v(l_t)] \\ \text{s.t.} \end{aligned} \quad (9.3.1)$$

$$a_{t+1} = (1 + r_t) a_t + w_t (1 - l_t) + \Pi_t^F + \Pi_t^I - c_t \quad (9.3.2)$$

$$a_0 = K_0 (1 + r_0^K - \delta) \text{ given}$$

a no-Ponzi-game condition like (6.3.7)

In each period, the household obtains utility from both consumption c_t and leisure l_t . Equation (9.3.2) is the budget constraint; more precisely, there is an infinite number of budget constraints, each on them linking one period to the following one. Π_t^F and Π_t^I represent the profits of the productive and investment firms respectively, which will turn out to be zero.

³There's a gray area with policies that are justified on the basis that people make the wrong choices. Should a benevolent social planner want to give people what they themselves would choose or what the social planner "knows" is best for them? This issue famously comes up in discussions of drug policy but also, for instance, in financial regulation. The answer involves a philosophical discussion that economists usually don't specialize in. Thaler and Sunstein (2008) discuss many policy issues related to this question.

The firm's problem is unchanged: it just maximizes profits period-by-period. The investment firm's problem is similar to (9.1.2), except that capital does not fully depreciate in one period, so a firm that invests at time t knows that it will get back depreciated capital at time $t + 1$ in addition to a rental. Therefore, the investment firm solves:

$$\Pi_t^I = \max_I \frac{r_{t+1}^K + 1 - \delta}{1 + r_{t+1}} [K_t(1 - \delta) + I] - [K_t(1 - \delta) + I] \quad (9.3.3)$$

Definition 9.2.

A competitive equilibrium consists of:

1. An allocation $\{c_t, l_t, I_t, K_{t+1}, L_t\}_{t=0}^\infty$.
2. Prices $\{w_t, r_t^K, r_t\}_{t=0}^\infty$.

such that:

1. $\{c_t, l_t\}_{t=0}^\infty$ solves the household's problem, taking prices as given.
2. L_t, K_t solve the firm's problem for every t , taking prices as given.
3. I_t solves the investment firm's problem for every t , taking prices as given.
4. Markets for goods and labor in each period clear:

(a) Goods:

$$\underbrace{F(K_t, L_t)}_{\text{GDP}} = \underbrace{c_t}_{\text{Consumption}} + \underbrace{I_t}_{\text{Investment}} \quad (9.3.4)$$

(b) Capital:

$$K_{t+1} = K_t(1 - \delta) + I_t \quad (9.3.5)$$

(c) Labor:

$$L_t + l_t = 1 \quad (9.3.6)$$

As you can see, there isn't much conceptual difference between how we define an equilibrium in a two-period model and in an infinite-period model. In fact, the description of how the equilibrium behaves is also very similar. The first order conditions for the household's problem are:

$$\frac{v'(l_t)}{u'(c_t)} = w_t \quad (9.3.7)$$

$$u'(c_t) = \beta(1 + r_{t+1}) u'(c_{t+1}) \quad (9.3.8)$$

For the firms problem:

$$F_K(K_t, L_t) - r_t^K = 0 \quad (9.3.9)$$

$$F_L(K_t, L_t) - w_t = 0 \quad (9.3.10)$$

And setting the NPV of investment to zero to prevent investment firms from choosing infinite investment:

$$r_{t+1} = r_{t+1}^K - \delta \quad (9.3.11)$$

which is just equation (4.4.12).

Replacing (9.3.10) into (9.3.7) we obtain:

$$\underbrace{\frac{v'(l_t)}{u'(c_t)}}_{\text{Marginal Rate of Substitution}} = \underbrace{F_L(K_t, L_t)}_{\text{Marginal Rate of Transformation}} \quad (9.3.12)$$

and replacing (9.3.11) and (9.3.9) into (9.3.8) and we obtain:

$$\underbrace{\frac{u'(c_t)}{\beta u'(c_{t+1})}}_{\text{Marginal Rate of Substitution}} = \underbrace{(1 + F_K(K_{t+1}, L_{t+1}) - \delta)}_{\text{Marginal Rate of Transformation}} \quad (9.3.13)$$

All of these equations have the same interpretation as in the two-period economy. The infinite-horizon model will just let us ask some additional questions, such as “what will the economy look like in the long run?” or “how does the economy react today to news about things that will happen in the distant future?”

Dynamics

When we studied the Solow model in Chapter 4, we concluded that an economy with a constant savings rate and a fixed labor supply would converge towards a steady state, where the capital stock and output were constant and investment was just enough to make up for depreciation. Now that we have a theory of what determines the savings rate, we can ask the same questions again: what will the economy look like in the long run? How will it behave in the meantime?

To keep things relatively simple and focus just on the consumption / investment problem, we are going to go back to the assumption that the labor supply is fixed, so instead of equation (9.1.12) we'll just have $L_t = 1$. Furthermore, we'll assume that the representative household has the utility function $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$, which we first introduced in Chapter 2. With this utility function and setting $L_t = 1$, equation (9.3.13) reduces to:

$$\frac{c_{t+1}}{c_t} = [\beta (1 + F_K(K_{t+1}, 1) - \delta)]^{\frac{1}{\sigma}} \quad (9.3.14)$$

Equation (9.3.14) gives us a relationship between the *rate of growth* of consumption between t and $t + 1$ and the *level* of the capital stock at time $t + 1$. If the capital stock is *low*, consumption should be *growing* over time. What is the economic logic that it represents? Suppose, for instance, that the capital stock at $t + 1$ is *low*. This implies that the marginal product of capital is *high*, due to diminishing marginal product; therefore the rental rate of capital will be *high*, by equation (9.1.9); therefore the interest rate will be *high*, by equation (9.1.11); therefore current consumption is expensive relative to future consumption, as we saw in Chapter 6;

therefore the household chooses to consume more in the *future* than in the *present*, by the Euler equation (9.1.8), which is exactly what a high rate of growth of consumption means.

In addition, we have the market clearing condition (9.1.3), which we can rewrite as:

$$K_{t+1} = (1 - \delta) K_t + F(K_t, 1) - c_t \quad (9.3.15)$$

This gives us a relationship between the future capital stock, the current capital stock, and consumption. The economic logic here is simple: more consumption implies less investment and therefore a lower capital stock in the following period.

We can represent equations (9.3.14) and (9.3.15) by means of a phase diagram. This is a graph that shows us, for each possible level of c and k , in what direction c and k are supposed to be moving if they are to satisfy (9.3.14) and (9.3.15). This is shown in Figure 9.3.1.

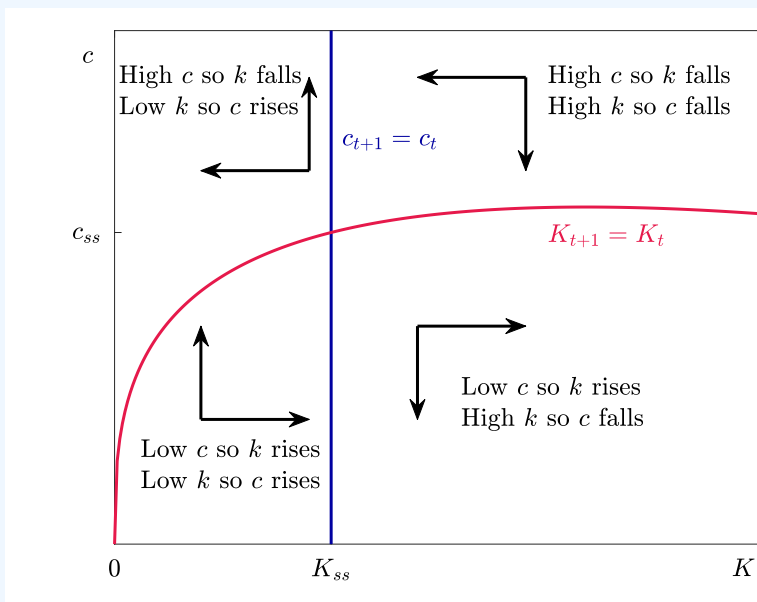


Fig. 9.3.1: Dynamics of the Neoclassical Growth Model

The vertical line represents equation:

$$\begin{aligned} \frac{c_{t+1}}{c_t} &= 1 \\ \Rightarrow [\beta (1 + F_K(K_{t+1}, 1) - \delta)]^{\frac{1}{\sigma}} &= 1 \end{aligned} \quad (9.3.16)$$

$$\Rightarrow F_K(K_{t+1}, 1) - \delta = \frac{1}{\beta} - 1 \quad (9.3.17)$$

so it tells us what level of capital is consistent with constant consumption. At any point to the left, the lower capital stock means a higher interest rate so consumption must be growing; at any point to the right, the higher capital stock means a lower interest rate so consumption must be falling. Only if K solves equation (9.3.17) is the interest rate exactly $\frac{1}{\beta} - 1$, which persuades the household to keep consumption constant over

time.

The curved line represents the equation:

$$\begin{aligned} K_{t+1} &= K_t \\ \Rightarrow (1 - \delta) K_t + F(K_t, 1) - c_t &= K_t \\ \Rightarrow c_t &= F(K_t, 1) - \delta K_t \end{aligned}$$

so it tells us, for each level of K , how much that household needs to consume in order to invest enough to exactly make up for depreciation, thus keeping the capital stock constant. For all the points above the curve, higher consumption implies that depreciation exceeds investment so the capital stock shrinks; for all the points below the curve, lower consumption implies that investment exceeds depreciation and the capital stock grows.

Mathematically, (9.3.14) and (9.3.15) are two difference equations in terms of K_t and c_t . If we knew the initial conditions K_0 and c_0 , we could trace out the entire path of both variables over time. The initial condition for K_0 is easy. We assumed it's exogenous so we just take as given its initial value. How about c_0 ? How much will the household consume in the initial period? Figure 9.3.2 shows, for two possible levels of the initial capital stock (K_0 and K'_0), the paths of K_t and c_t that result from different possible values of c_0 .

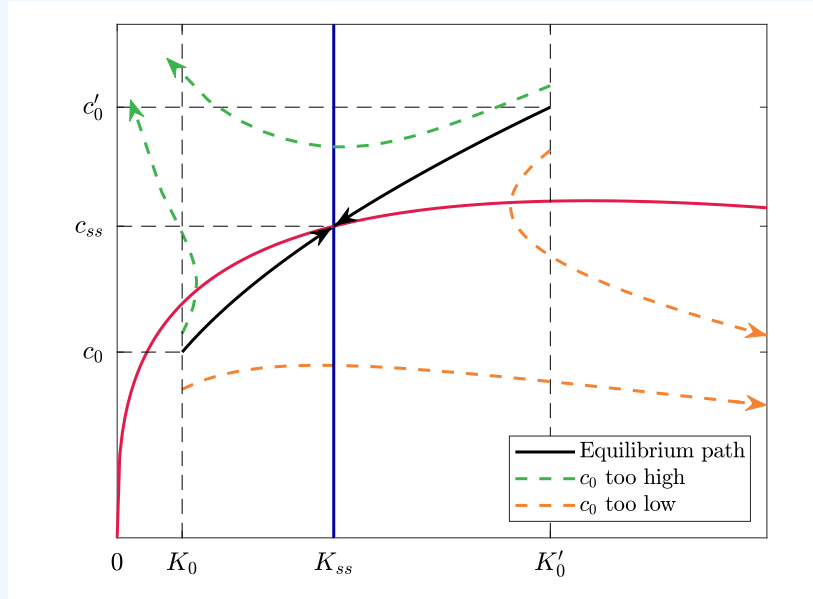


Fig. 9.3.2: Paths for the economy dictated by equations (9.3.14) and (9.3.15), starting from different levels of c_0 and K_0 .

In each case, the higher path has c_0 too high. Starting from this level of c_0 , the Euler equation (9.3.14) dictates path of consumption that becomes ever-increasing, but there is not enough output so the economy starts to deplete the capital stock, eventually depleting it completely. In the lower paths, c_0 is too low. The economy accumulates more and more capital over time and after some time the Euler equation starts to dictate falling consumption. Eventually, all the output is being invested and consumption falls to zero. The only levels of c_0 that are consistent with optimality are the ones that gives rise to the middle paths. Here both c_t and K_t

converge to c_{ss} , K_{ss} . Hence this economy, just like the Solow economy with an exogenous savings rate, has a steady state.

In the theory of difference equations, the paths that lead to the steady state are known as “saddle paths” and the steady state is known as a “saddle point”. Starting from any K_0 , there is a unique level of c_0 such that the dynamics implied by equations (9.3.14) and (9.3.15) lead towards the steady state. Furthermore, once c_t is on the saddle path, it will stay on the saddle path, so the economy will always be on this path.

The Golden Rule

When we looked at the Solow model we defined a concept called the Golden Rule. This described the level of capital K_{gr} (and the savings rate needed to attain it) such that steady state consumption is maximized. We saw that, depending on the savings rate, an economy could end up with either more or less capital than prescribed by the Golden Rule. Now that we have a theory of the savings rate we can ask how the model predicts that K_{ss} and K_{gr} will compare. Will the economy accumulate more or less capital than prescribed by the Golden Rule? Recall from Chapter 4 that equation (4.3.4) says K_{gr} satisfies:⁴

$$F_K(K_{gr}, 1) = \delta$$

Instead, with endogenous savings, the steady state level of capital satisfies (9.3.17). Rearranging, we have

$$\begin{aligned} F_K(K_{ss}, 1) &= \frac{1}{\beta} - 1 + \delta \\ &> \delta && \text{(as long as the household is impatient so } \beta < 1) \\ &= F_K(K_{gr}, 1) \\ \Rightarrow K_{ss} &< K_{gr} \end{aligned}$$

Therefore this economy will *not* attain the Golden Rule level of capital. It will remain below this.

What do we make of this? The Golden Rule seemed like a pretty desirable outcome, and the First Welfare Theorem says that the equilibrium maximizes the household’s utility: why doesn’t the social planner implement the Golden Rule? The answer comes from the household’s impatience. The Golden Rule maximizes consumption *in the long run*. An impatient household cares about the short run as well as the long run. It would rather consume a little bit more in the present even if it means a lower level of consumption later. Note that mathematically, $K_{ss} \rightarrow K_{gr}$ if $\beta \rightarrow 1$, so as households become very patient the economy indeed comes closer to the Golden Rule. Note also that the argument does not depend on starting with a low level of capital. If the economy were to start at K_{gr} , the household would choose to invest less than required to maintain the capital stock, consuming more than c_{gr} in the short run at the expense of lower consumption later.

Figure (9.3.3) shows how the steady state compares to the Golden Rule. The graph shows, for each value of K , the level of consumption that is consistent with maintaining a constant capital stock equal to K , i.e. $c = F(K, 1) - \delta K$. If the economy were to maintain K_{gr} , then it could sustain a level of consumption $c_{gr} > c_{ss}$. However, attaining and maintaining such a high capital stock requires sacrificing too much present

⁴In Chapter 4 we had population growth. Here we are setting $n = 0$ but the argument works regardless.

consumption for future consumption and the household is better off with the equilibrium that converges to K_{ss} .

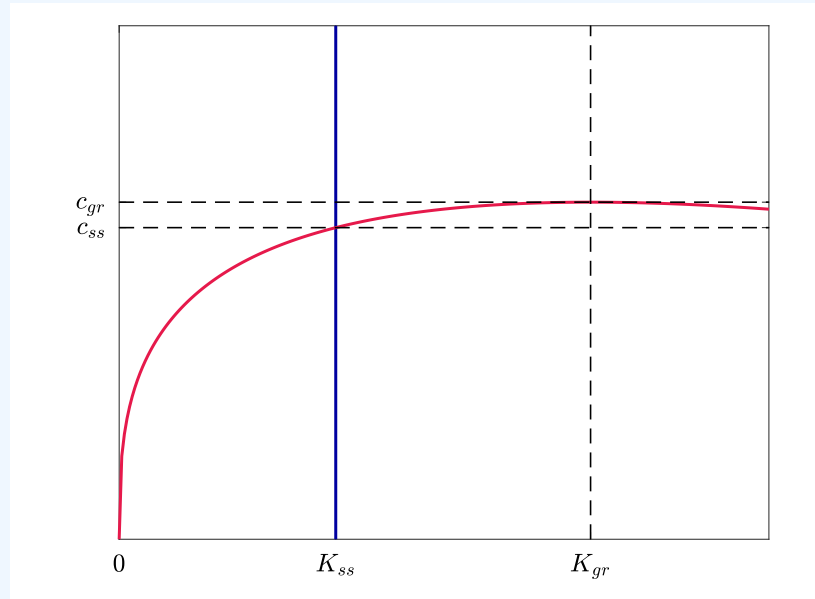


Fig. 9.3.3: How the steady state compares to the Golden Rule.

Anticipation Effects

Thinking in terms of general equilibrium can be useful for thinking about how anticipation of things that will happen in the future can affect decisions in the present. Let's consider an example. Suppose the economy is in steady state and suddenly everyone anticipates that a technological breakthrough will lead to a change in the production function from $F(K, 1)$ to $AF(K, 1)$ starting in year T , where $A > 1$. Figure 9.3.4 shows what the effect of this would be.

Once it happens, the technological improvement shifts the $c_t = c_{t+1}$ line to the right: higher A means that it takes higher K to have $AF_K(K, 1) = \frac{1}{\beta} - 1 + \delta$ (which is the condition for consumption stay constant). In addition, the $K_t = K_{t+1}$ curve shifts up: higher productivity means it is possible to afford more and still maintain the capital stock. Once period T arrives, we can apply the analysis we did for the constant-technology case. From T onward, the economy must be on the new saddle path that leads to the new steady state. But before period T , the dynamics of capital and consumption are still governed by the old technology. As illustrated in the figure, the initial level of consumption must be such that, by the time period T arrives, the dynamics under the *old* technology lead to the saddle path of the *new* technology.

In economic terms, what happens is that anticipation of a technological improvement leads to higher consumption through a wealth effect. Since technology has not improved yet, higher consumption implies that the economy is investing less than is necessary to maintain the old capital stock, so the capital stock begins to shrink. A lower capital stock means that the marginal product of capital is higher and therefore the interest rate is higher than $\frac{1}{\beta} - 1$. This means that the household chooses a rising path for consumption. Hence, in

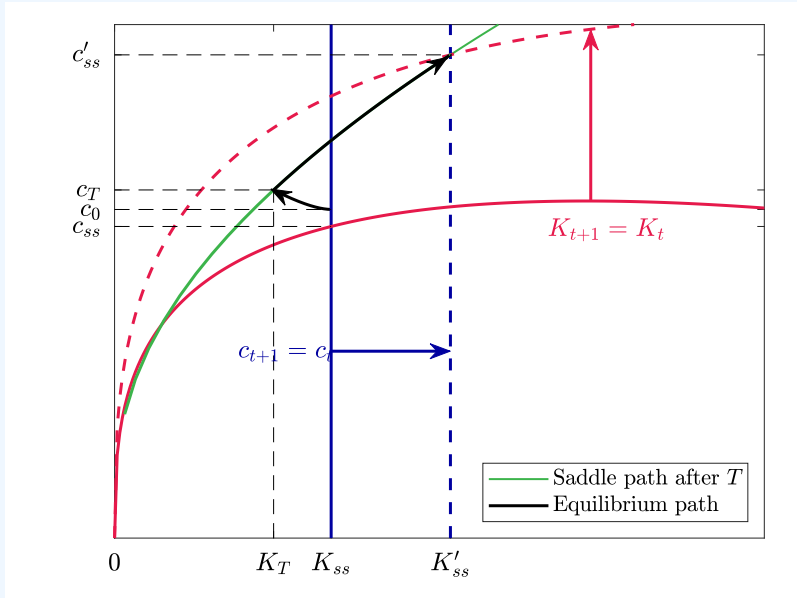


Fig. 9.3.4: An anticipated technological improvement.

anticipation of the technological improvement, consumption first jumps from c_{ss} to c_0 and then gradually rises over time. When period T arrives, the capital stock has shrunk to K_T and consumption has reached c_T , so the economy is exactly in the saddle path that will lead it to the new steady state K'_{ss}, c'_{ss} .

Exercises

9.1 The First Welfare Theorem

Consider the following policies. In each case, explain whether, in your view, the policy is justified and why.

- Setting a maximum price that electrical utilities may charge.
- Mandatory vaccinations.
- Workplace safety standards.
- Rent control.
- A 35-hours-per-week limit on working hours.
- Banning grocery stores from opening on Sundays.
- Banning self-service at gas stations.

9.2 Storage

Suppose the production function is the following:

$$F(K, L) = K$$

and the rate of depreciation is $\delta = 1$.

- Why is this production function called “storage”?
- What will be the real interest rate and the wage in this economy?
- With standard preferences, will consumption increase over time, remain constant over time or decrease over time?

9.3 Baumol’s Cost Disease

The representative household consumes two different goods: x (clothes) and y (performances by string quartets). Its preferences are given by:⁵

$$u(x, y) = \left(\alpha^{\frac{1}{\epsilon}} x^{\frac{\epsilon-1}{\epsilon}} + (1 - \alpha)^{\frac{1}{\epsilon}} y^{\frac{\epsilon-1}{\epsilon}} \right)^{\frac{\epsilon}{\epsilon-1}}$$

Each of the two goods is produced using only labor. If L_x units of labor work in producing clothes, output of clothes is $x = A_x L_x$; if L_y units of labor play in string quartets, output is $y = A_y L_y$. The household supplies one unit of labor inelastically and is indifferent as to how much it supplies to each industry. The price of clothes is denoted p_x and the price of a quartet performance is p_y . The wage rate is w .

- Use a software like Matlab or Excel to plot two sets of indifference curves for this utility function. Set $\alpha = 0.5$ in both cases, $\epsilon = 0.5$ in one of them and $\epsilon = 2$ in the other.
- Set up the problem of a household that obtains income w from supplying one unit of labor inelastically and has to decide how much to consume each of the two goods. Show that the household will consume:

$$x = \frac{w}{p} \alpha \left(\frac{p_x}{p} \right)^{-\epsilon}$$

$$y = \frac{w}{p} (1 - \alpha) \left(\frac{p_y}{p} \right)^{-\epsilon}$$

where $p \equiv (\alpha p_x^{1-\epsilon} + (1 - \alpha) p_y^{1-\epsilon})^{\frac{1}{1-\epsilon}}$. [This takes a bit of work. If you want, you can look up the derivation on the book website].

- Set of the problem of a firm in each of the two industries that needs to decide how much labor to hire to maximize profits. What values of $\frac{p_x}{w}$ and $\frac{p_y}{w}$ are consistent with each industry hiring a positive but not infinite amount of labor?
- What will be the price of string quartet performances relative to clothes $\frac{p_y}{p_x}$?
- Suppose that over time A_x rises and A_y stays the same. What does this mean? Do you find this plausible? What will be the effect on the relative price of string quartets? Explain.
- Use your answers to parts (b) and (d) to compute the relative quantities of each of the two goods that will be consumed, i.e. solve for the equilibrium level of $\frac{y}{x}$.

⁵This is known as a “constant elasticity of substitution” utility function. The parameter ϵ represents the elasticity of substitution between goods x and y : a higher number means that they are close to perfect substitutes. For reference, when $\epsilon \rightarrow 1$, these preferences converge to the Cobb-Douglas case $u(x, y) = x^\alpha y^{1-\alpha}$.

- (g) Use your answer to part (f) to compute the relative allocation of labor across the two industries, i.e. solve for the equilibrium level of $\frac{L_y}{L_x}$.
- (h) Suppose that over time A_x rises and A_y stays the same. What happens to the quantity of string quartet performances over time? How does the answer depend on ϵ ? Explain.

9.4 Pro-Worker Policies

Consider the following one-period economy. The production function is $F(K, L)$. The representative household owns all the capital stock K , which is exogenously given. The household's preferences are $u(c, l)$ where, as usual, $l = 1 - L$. The labor and capital markets are perfectly competitive.

Denote by L^* the amount of labor that households supply in a competitive equilibrium with no government intervention. Now suppose that the government dictates a new law that prohibits the household from working more than $L^* - \varepsilon$, where ε is a small positive number.

- (a) Show that the policy will increase wages and lower the rental rate of capital.
- (b) Show that the policy will make the representative household worse off.
- (c) Now suppose that there are two representative households in the economy. Household A owns the capital stock and does not work. Its preferences are given by $u(c)$. Household B has preferences $u(c, l)$ and does not own any capital. Suppose the government enacts the same policy as before (i.e. it prohibits household B from working more than $L^* - \varepsilon$, where L^* is the amount it works under a no-intervention equilibrium). Show that this policy makes household A worse off and household B better off.

9.5 Differences in Preferences

There are two countries: Industria and Lethargia. Both countries have a population of 1 and identical production functions:

$$Y = AL$$

where L is the total amount of time spent in market labor.

Preferences are given by:

$$u(c, l) = \log(c) + \theta \log(l) \tag{9.3.18}$$

where c is consumption and l is leisure. Within each country, everyone is identical but θ is different for residents of the two countries. Denote the two values by θ^I and θ^L and assume $\theta^L > \theta^I$.

Both countries run free market economies, with perfectly competitive labor markets.

- (a) What is the wage in each country?
- (b) Set up the problem of a representative household that has to decide how to divide its time (normalized to a total of 1) between market work and leisure.
- (c) What fraction of its time will the representative household spend on market work in each country?
- (d) Which country has higher GDP?

(e) Suppose a researcher does the following:

- i. assumes preferences are the same across the world, with functional form (9.3.18),
- ii. estimates θ using data from Lethargia,
- iii. gathers data on consumption and leisure in each country,
- iv. computes λ : the relative welfare of residents of Industria, taking Lethargia as a benchmark with $\lambda = 1$ the way Jones and Klenow took the US as a benchmark.

Find an expression for λ . Which country would the researcher conclude has a higher standard of living? Explain.

9.6 Capital Income Taxes

An economy has the production function

$$Y = K^\alpha L^{1-\alpha}$$

It is populated by two types of households: workers and capitalists.

Workers supply a total of L units of labor inelastically and consume all their income in every period, so they don't really make any decisions. Their consumption in any given period is given by:

$$c_t = w_t L + T_t$$

where w_t is the wage and T_t is a transfer they get from the government.

Capitalists have preferences given by:

$$\sum_{t=0}^{\infty} \beta^t u(c_t)$$

They do not work: they get their income from capital, and it is taxed at a rate τ . However, they can choose how much to consume and how much to save in the standard way. Their budget constraint is:

$$K_{t+1} = -c_t + (1 + (1 - \tau) [r_t^K - \delta]) K_t$$

where K_t is the capital stock in period t , δ is the rate of depreciation, τ is the tax rate and r_t^K is the rental rate of capital (which each capitalist household takes as given but depends on the total capital stock in the usual way).

What does this budget constraint mean? For each unit of capital the capitalist has, it obtains a rental. The government taxes this rental (net of depreciation) at a rate τ . Assume that the government taxes interest income in the same way as it taxes income from renting capital, so that equation (9.1.11) holds.

- (a) Set up the maximization problem of a capitalist household.
- (b) From the first order conditions, find an after-tax version of the Euler equation. (You can skip steps if you want)
- (c) If in the long run this economy reaches a steady state in which consumption of the capitalists is constant, what are the pre-tax and after-tax interest rates in this steady state?

- (d) If the economy is in a steady state, what is the rental rate of capital? How does it depend on τ ?
- (e) If the economy is in a steady state, what is the capital stock?
- (f) How much revenue does the government collect from the capital-income tax in steady state?
- (g) What is the level of wages in steady state?
- (h) Suppose that the government uses all the revenue from the capital-income tax to finance transfers to the workers. What level of consumption do workers attain in steady state? How does it depend on τ ? Explain.

9.7 Getting Old

An economy has two types of households: young and old. Let μ be the fraction of households that are young. We hold the total population constant and consider the effects of higher or lower μ (which could be the result of differences in fertility and mortality). Each young household supplies x units of labor inelastically, so the total amount of young labor is $L^Y = \mu x$; old households supply z units of labor, also inelastically, so the total amount of old labor is $L^O = (1 - \mu)z$. Assume $x > z$.

The production function is:

$$Y = K^\alpha L^{1-\alpha}$$

where K is the capital stock and L is the total amount of labor: $L \equiv L^Y + L^O$.

- (a) Holding K constant, how do GDP, wages and the rental rate of capital depend on μ ?
- (b) How will the steady-state capital stock compare between two economies with different levels of μ ? Explain.
- (c) Now imagine instead that the economy doesn't use capital and the production function is:

$$Y = (L^Y)^\gamma (L^O)^{1-\gamma}$$

How do the wage levels of young and old workers depend on μ ?

9.8 A New Technology

Consider a one-period economy where the representative worker has preferences:

$$\log(c) + \theta \log(l)$$

where c is consumption and l is leisure. The worker must choose how to divide its time between leisure l and market work L , so:

$$l + L = 1$$

The labor market is perfectly competitive. The equilibrium wage is denoted w .

- (a) Set up the problem of the representative worker, who must decide how to allocate his time. Assume the worker has no source of income other than its earnings from labor. Solve for the household's choice of L .

The production technology uses labor and capital to produce goods according to:

$$Y = K^\alpha L^{1-\alpha}$$

where K is the capital stock, which is exogenously given. The capital-rental market is also perfectly competitive. The equilibrium rental rate of capital is denoted r^K .

- (b) Set up the problem of the representative firm, which must choose capital and labor inputs to maximize profits. Derive first order conditions and find the level of w and r^K such that markets clear.
- (c) How do w and r^K depend on K and θ ? Explain.
- (d) Suppose a team of scientists develops a new technology for producing goods. Using this new technology, it's possible to produce goods according to:

$$Y = AK$$

where A is a parameter. The scientists tried out the technology at a very small scale, so anything they did was too small to affect equilibrium prices. Prove that the new technology turned out to be profitable if and only if $K > \bar{K}$, where $\bar{K}$ is some number. Find an expression for $\bar{K}$. How does $\bar{K}$ depend on θ ? Explain.

9.9 An Oil-Producing Economy

Suppose that the production function is given by $F(K, L) = K^\alpha L^{1-\alpha} + e$. GDP is the sum of regular output, which is produced using capital and labor, and oil extraction e , which is exogenous. The representative household has preferences:

$$\sum_{t=0}^{\infty} \beta^t \log(c_t)$$

and supplies $L = 1$ units of labor inelastically. Capital depreciates at rate δ .

- (a) On the same graph, draw two phase diagrams, one for $e = 0$ and one for $e = \bar{e} > 0$ and label the steady state in each case.
- (b) Suppose that the economy starts at $t = 0$ at the $\bar{e}$ steady state but everyone suddenly realizes that oil will run out at time T , so that:

$$e_t = \begin{cases} \bar{e} & \text{if } t < T \\ 0 & \text{if } t \geq T \end{cases}$$

Using the phase diagrams, plot the evolution of K_t and c_t from $t = 0$ until $t = T$ and from $t = T$ onwards. Describe in words why consumption and investment evolve the way they do, and what happens to wages and the interest rate.

9.10 Wizards and Cows

The Calevingians have two basic economic activities: gathering fruit and raising cows.

Wild fruit grows in the Western part of the kingdom. The Calevingians have not mastered the art of growing fruit trees, so they have no control over how much fruit is available. They just visit the fruit trees every week and pick whatever fruit is ripe. Luckily, the kingdom includes fruits with different seasonal patterns so there is ripe fruit more or less evenly throughout the year. Not all years are the same, however. Some years are warm and fruit is plentiful, while others are colder and result in less fruit. The Calevingians have complete trust in their chief wizard, who gives them a weather forecast (and thus a fruit forecast) with a horizon of a few years.

Cows graze in pastures in the Eastern part of the kingdom. The grass on which cows feed grows just as well in cold and warm weather. Each family owns a plot of land and keeps their own cows there. Cows are raised primarily for meat; the Calevingians have not learned to milk them. One of the main decisions that Calevingians need to make is how many of their cows to slaughter for meat each year and how many (including newborn calves) to keep fattening from one year to the next. They understand that the more cows they keep in their plot of land, the less grass each of them will have available for grazing.

- (a) Write down mathematically the economic decision problem faced by a Calevingian household. Explain the meaning of each equation you write down.
- (b) Derive first-order conditions (you may skip steps if you want).
- (c) Suppose the chief wizard announces that the next few years will be cold and therefore yield less fruit. How do households react to this announcement?
- (d) Despite their rather simple economy, the Calevingians have a fairly sophisticated legal system; the concept of a sale, a loan and an interest rate are well-established and contracts are enforced very effectively. It's possible, for instance, to borrow in order to buy more cows or, conversely, to sell one's cows and lend the proceeds of the sale to someone else. What happens to interest rates when the wizard announces the upcoming cold weather?

9.11 Patience and Investment

Suppose the production function is $F(K, L) = K^\alpha L^{1-\alpha}$, and the labor supply is exogenous and equal to $L = 1$. The representative household's preferences are given by:

$$\sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\sigma}}{1-\sigma}$$

and the depreciation rate is δ .

- (a) Use equation (9.3.17) to find an expression for the level of capital that this economy will have in the steady state.
- (b) Find an expression for the level of GDP that the economy will have in the steady state.
- (c) Find an expression for the steady-state investment-to-GDP ratio. How does it depend on β ? Explain.

9.12 Optimal vs Fixed Savings Rates

Suppose the economy is as in Exercise 9.11, with $\alpha = 0.4$, $\beta = 0.95$, $\delta = 0.08$ and $\sigma = 2$. The economy starts with an initial capital stock $K_0 = 2$. We are going to compare how this economy behaves under a fixed investment rate and under an optimally-chosen investment rate.

- (a) First suppose that, like we assumed in Chapter 4, the investment rate is exogenous and constant, and equal to the steady-state investment-to-GDP ratio that you found in Exercise 9.11. Compute GDP, consumption, investment and the capital stock for the first 150 years of this economy.
- (b) Now suppose that consumption and investment are chosen in accordance with (9.3.13). Find the level of initial consumption that is consistent with the system of equations (9.3.15) - (9.3.13) and the economy converging to a steady state. (You can find a Matlab code that will compute this at the book website). Compute GDP, consumption, investment and the capital stock for the first 150 years of this economy.
- (c) Plot the consumption paths of the two economies over the first 150 years on the same graph.
- (d) Replace the computed paths of consumption from parts (a) and (b) into the utility function to compute how much utility the representative household will obtain in the first 150 years of each economy.
- (e) By what fraction would one have to reduce consumption in the optimal-savings economy for the household to get the same utility as in the exogenous-savings economy?

9.13 The Enclosure Acts

The Enclosure Acts were a series of laws that converted common property into private property in Great Britain starting around the XVII century. This exercise asks you to think about their macroeconomic effects.

Consider the following one-period economy.

- There are two sectors:
 - Agriculture produces output according to:

$$Y_A = N^\alpha L_A^{1-\alpha}$$

where Y_A is agricultural output, N is natural resources (land) and L_A is labor dedicated to agriculture.

- Industry produces output according to:

$$Y_I = K^\alpha L_I^{1-\alpha}$$

where Y_I is industrial output, K is capital and L_I is labor dedicated to industry.

- For simplicity, assume that both sectors produce the same good (or that both goods are perfect substitutes), so GDP is $Y = Y_A + Y_I$.

- The industrial sector operates as a competitive market, with a wage rate w and a rental rate of capital r^K . The capital stock is exogenously given and is owned by capitalists, who do not work.
 - Agriculture is organized as follows:
 - no one owns the land and anyone can use it,
 - once all the agricultural output is produced, it gets shared equally among agricultural workers.
 - The amount of land available and the capital stock are exogenously given.
 - Each worker supplies one unit of labor. The total number of workers is L .
 - Each worker freely decides whether to work in industry or in agriculture:
 - if the worker chooses industry, he gets a wage w ,
 - if the worker chooses agriculture, he gets a share of total agricultural output
- (a) Find an expression for the ratio $\frac{L_A}{L_I}$ that would lead to maximum GDP.
- (b) If there are L_A workers in agriculture, what is the income of a worker who chooses agriculture?
- (c) If there are L_I workers in industry, what is the income of a worker who chooses industry?
- (d) What is the ratio $\frac{L_A}{L_I}$ that would make workers indifferent between choosing agriculture and industry? Argue that this is the allocation of labor that the economy will have. How does it compare to the allocation that would maximize GDP? Explain.
- (e) Suppose there is an Enclosure Act that does the following:
- It splits up the land equally among agricultural workers, giving each of them ownership of a specific piece of land.
 - It establishes a competitive market for renting land and hiring agricultural workers.

After the reform:

- i. Which of the sectors will increase the number of workers it employs? Explain.
 - ii. What will happen to GDP? Explain.
 - iii. What will happen to industrial wages? Explain.
 - iv. What will happen to the rental rate on industrial capital? Explain.
- (f) How would the answers to part (e) change if instead of being split up equally the land was given to only some of the agricultural workers? Why?